Generalized Linear Models

Applied Linear Models I • Lecture 16 of 16

Applied Linear Models I

Today (then the final)

Outcomes that break the linear model – binary, counts, rates:

- **logistic** regression and odds ratios;
- logit vs probit;
- **Poisson** regression, rate ratios, offsets;
- overdispersion (quasi-Poisson, negative binomial).

The final covers the whole course, emphasizing Lectures 9–16.

The GLM recipe

Keep the linear predictor $\eta = \mathsf{X}\boldsymbol{\beta}$; add two ideas:

- **random component**: response distribution (binomial, Poisson, ...);
- **link** $g(\mu) = \eta$: maps a constrained mean to all of $\mathbb{R}$.

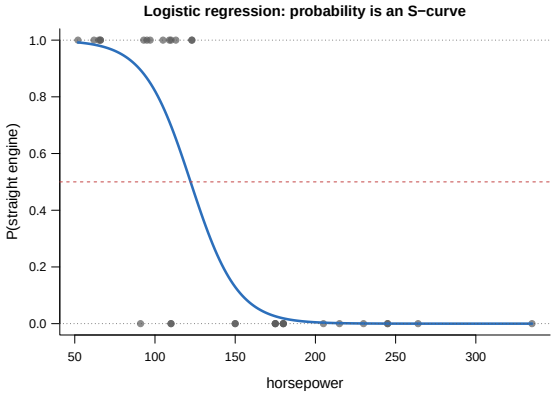
Fit by **IRLS** (maximum likelihood) – same engine as robust regression.

The linear model is the GLM with a *normal* response and the *identity* link.

Logistic regression

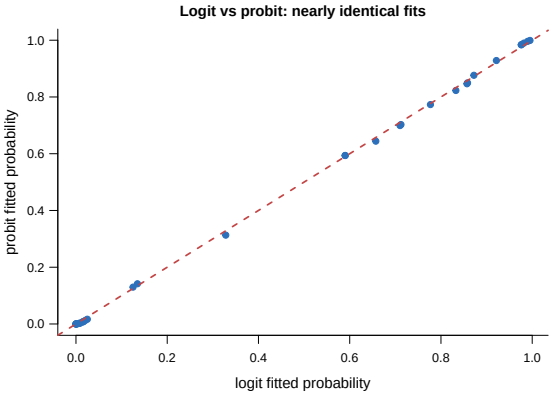
$$\log \frac{p}{1-p} = \mathrm{x}^\top \boldsymbol{\beta} \quad \Longleftrightarrow \quad p = \frac{1}{1 + e^{-\mathrm{x}^\top \boldsymbol{\beta}}}.$$

- Maps η into $(0, 1)$ – the **S-curve**.
- $e^{\beta_j} =$ **odds ratio**.
- Effect on *probability* is nonlinear.



Logit vs probit

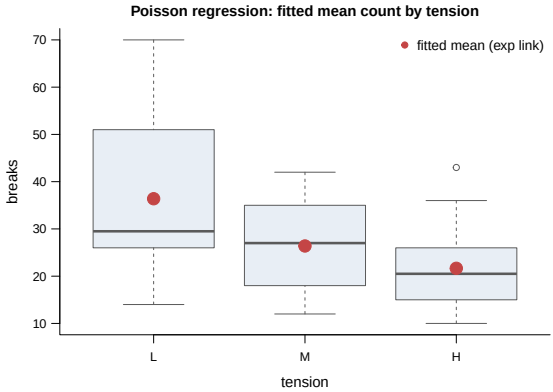
- Probit link: $\Phi^{-1}(p) = \mathbf{x}^\top \boldsymbol{\beta}$.
- Fitted probabilities **nearly identical** (corr > 0.99).
- Coefficients differ by ≈ 1.6–$1.8\times$ (scale).
- Logit preferred for odds ratios; probit common in econometrics.



Poisson regression

$$\log \mu = \mathrm{x}^\top \boldsymbol{\beta} \quad \Longleftrightarrow \quad \mu = e^{\mathrm{x}^\top \boldsymbol{\beta}}.$$

- Log link $\Rightarrow \mu > 0$, **multiplicative** effects.
- e^{β_j} = **rate ratio**.
- **Offset** log(exposure) for rates.



`warpbreaks`: high tension $\times 0.60$ on the break rate.

Overdispersion

Poisson assumes $\text{Var}(y) = \mu$. Real counts often have $\text{Var}(y) > \mu$. `warpbreaks` dispersion ≈ 4.3 – ignoring it makes SEs **too small**.

Remedies

- **quasi-Poisson**: inflate SEs by $\sqrt{\phi}$ (here $\approx 2\times$).
- **negative binomial**: model the extra variance, $\text{Var}(y) = \mu + \mu^2/\theta$.

Same concern as robust SEs (L6) and WLS/GLS (L14) – now for counts.

GLMs unify the course

- Design matrices, linear predictor, nested tests (now deviance/LRT), diagnostics, AIC, validation, regularization – all extend.
- Deviance generalizes SSE; IRLS generalizes the normal equations.
- The link is the only genuinely new idea.

Course complete

The final covers the whole syllabus (emphasis L9–16); the capstones integrate diagnostics, remediation, validation, and GLMs.

Summary

- GLM = random component + linear predictor + link, via IRLS.
- Logistic: logit link, e^{β} odds ratio, S-curve.
- Poisson: log link, e^{β} rate ratio, offsets for exposure.
- Overdispersion inflates true SEs – quasi-Poisson or negative binomial.
- The linear model is one GLM among many.

Thank you – end of Applied Linear Models I.